

Zongze Li

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SKILL SETS

Simulation and Modeling

Computational Fluid Dynamics (CFD), Finite Element Analysis/Method (FEA/FEM), Computer-Aided Engineering (CAE), Engineering Analysis, Verification & Validation, Fluid-Structure Interaction (FSI), Lattice Boltzmann Method (LBM), Multi-Physics Simulation (Thermal-Fluid-Solid Coupling), Numerical Method Stabilization, ODE/PDE Discretization, End-to-end simulation workflow (CAD to Simulation to Analysis), Parametric and Design-Space Studies, Post-Processing Automation

Simulation Tools/Software

ANSYS Fluent, COMSOL, Palabos, SolidWorks, AutoCAD, Blender, Fusion 360, ImageJ, MeshLab, MATLAB, Paraview

Programming

Algorithm Development, C++, High Performance Computing (HPC), HTML, JAVA, Linux, Microsoft Visual Studio, Python, Message Passing Interface (MPI)

Laboratory

2D Laser Cutting, 3D Printing, CNC Milling, Drilling, Milling, Particle Image Velocimetry (PIV)

SUMMARY

Simulation and Modeling Engineer | Ph.D. in Mechanical Engineering

7+ years of experience developing end-to-end simulation workflows for engineering design and decision-making. Strong background in multi-physics modeling (CFD, FSI, etc.) with experience in ANSYS, COMSOL, open-source frameworks, and customized C++ code package development for simulation.

Experienced in numerical modeling, solver development, high-performance parallel computing (MPI/Linux), simulation automation, validation, and post-processing visualization for engineering problems. Fast learner with strong multidisciplinary collaboration skills and growing interest in AI-assisted simulation methods such as Physics-Informed Neural Networks (PINNs).

PROJECTS

Multi-Physics Simulation of Marine Propulsion Performance (2022-2025, [Click Here](#) to See the Work)

The goal of this project was to analyze a marine organism propulsion performance and support bio-inspired underwater robotic design through multi-physics simulation. The workflow was organized into the following procedures:

1. **Real-world physics to numerical model conversion:** Converted physical problems into feasible numerical simulation models through simulation domain selection, governing ODE/PDE formulation, geometry simplification, and boundary condition design.
2. **Customized simulation package development:** Developed customized C++ simulation packages based on the open-source Palabos framework to achieve fluid-structure interaction (FSI) simulations, including ODE/PDE discretization, numerical stability improvement, MPI-based parallel solver development in Linux/HPC environments, workflow automation, solver validation, and user-friendly interface design.
3. **Simulation post-processing and engineering analysis:** Conducted parametric studies, validated simulations against experimental data, and developed visualization/post-processing pipelines to communicate simulation results to non-simulation engineers and designers.

The workflow was designed to be highly automated and **reusable for future scalable engineering workflows**. This project was **selected as a featured work by the American Institute of Physics (AIP)**, [click here](#) to see the News.

Tools used (number denotes workflow procedure number): MATLAB (1,3), Python (1,3), Blender (1), MeshLab (1), ParaView (1,3), DLTdv7 (1), C++ (2,3), Palabos (2), MPI (2), Linux (2), HPC (2), HTML (2), ANSYS Fluent (3), VS Code (1,2,3)

CFD Modeling of Human Pulsatile Aortic Flow (2019-2022, [Click Here](#) to See the Work)

The goal of this project was to analyze blood flow inside the human aorta that are difficult to directly measure using current medical equipment, providing simulation-assisted insights for clinical decision-making. The overall workflow was similar to the previous project, with the following key technical differences:

- **3D-0D multi-scale coupling:** Developed customized C++ code package to couple a 0D lumped parameter model with a 3D CFD solver, enabling customized realistic boundary conditions beyond default simplified approaches.
- **Explicit ODE/PDE solver development:** Discretized governing ODE/PDE systems into numerical forms and utilized efficient explicit numerical method for large-scale simulations.
- **Numerical stabilization:** Improved numerical stability and robustness by implementing Runge-Kutta stabilization methods.
- **Verification and validation with experimental and commercial software data:** Validated simulation results against experimental data and benchmarked results with commercial software including ANSYS and COMSOL to improve simulation credibility and reliability.

The simulation achieved high accuracy (less than 5% error compared with target values), and the published work **has been cited and adopted by multiple research groups**.

Novel Boundary Condition Development Based on Open-Source Solver Framework – Palabos (2021-Present)

The goal of this project is to develop robust and accurate boundary condition algorithms for the open-source CFD framework Palabos, addressing limitations in existing boundary treatment methods for complex fluid simulations. Key contributions include:

- **Numerical method development:** Developed two novel boundary condition algorithms for the open-source solver framework to improve simulation accuracy, numerical stability, and robustness.
- **Linear algebra development:** Developed customized C++ package for solving linear algebra systems of the form $Ax = B$ using Gaussian elimination methods.
- **High-performance computing (HPC):** Maintained scalable computational performance by integrating MPI into the simulation workflow within Linux/HPC environments.

Thesis and Course Projects

Design and Testing of Experimental Langmuir Turbulence Facilities (ANSYS FLUENT, 2018-2019)

Thermal Environment Analysis and Optimization in PC Chassis (ANSYS FLUENT, Jun/2015)

CFD Modeling of Air Circulation by Ceiling Fan (ANSYS FLUENT, Apr/2021)

6-DOF Robotic Arm Design (AutoCAD & MATLAB, Apr/2018)

EDUCATION

Ph.D. in Mechanical Engineering at University of South Florida, Tampa, FL (2019-Present, GPA: 3.96)

M.S. in Mechanical Engineering at University of South Florida, Tampa, FL (2017-2019, GPA: 3.95)

B.Eng. at University of Shanghai for Science and Technology, Shanghai, China (2011-2015, GPA: 3.21)

EXPERIENCE

Peer Reviewer for [Physics of Fluids](#) and [CFD Letters](#) (2024-Present)

[Publication](#) in Physics of Fluids (Apr/2026), featured in [AIP Scilight](#) for research significance

[Publication](#) in Computers & Fluids (Jun/2023)

Invited Speaker, U.S. Education without Borders (Mar/2022)

Conference Presenter: APSDFD ([Nov/2021](#) [Nov/2022](#)); FFS (May/2022, 2023, 2024, 2025)

Conference Poster: DCB-DVM ([Nov/2021](#))

Research and Teaching Assistant, University of South Florida (2019-Present)